

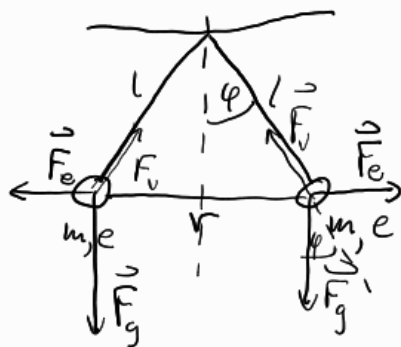
29/1

$$l = 12 \text{ cm}$$

$$m = 1 \text{ g}$$

$$e = 10^{-6} \text{ As}$$

$$2\varphi = ?$$



$$r = 2 \cdot l \cdot \sin \varphi$$

$$\tan \varphi = \frac{F_e}{F_g} = \frac{e^2}{mg \cdot 4\pi \epsilon_0 (2l \sin \varphi)^2}$$

$$\tan \varphi = \frac{c}{\sin^2 \varphi}, \quad c = \frac{e^2}{mg \cdot 16\pi \epsilon_0 l^2}$$

$$c = 15,9$$

$$\tan \varphi \geq c$$

$$\tan \varphi' = c \rightarrow \varphi' = 86,4^\circ$$

je ok, ker je
 $\sin(86,4^\circ) \approx 1$

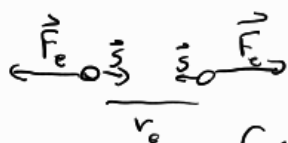
29/2

$$e = 3 \cdot 10^{-7} \text{ As}$$

$$r_0 = 10 \text{ cm}$$

$$r_1 = 3 \text{ cm}$$

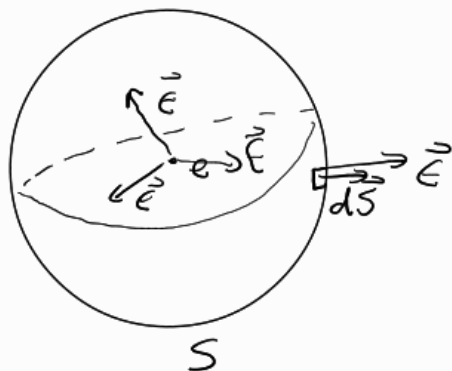
$$A = ?$$



$$A = \int_{r_0}^{r_1} \vec{F} \cdot d\vec{s} = - \int_{r_0}^{r_1} \frac{e^2}{4\pi \epsilon_0 r^2} dr = \frac{e^2}{4\pi \epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_0} \right) \approx 0,019 \text{ J}$$

$$W_{p,e} = \frac{e^2}{4\pi \epsilon_0 r}$$

Gaussov izrek

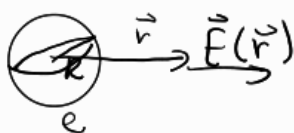


integral po zaključeni površini

$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = e$$

13.1.3

$$\rho_e = \text{konst.} = \frac{e}{4\pi R^3}$$



$E:$

$r > R:$

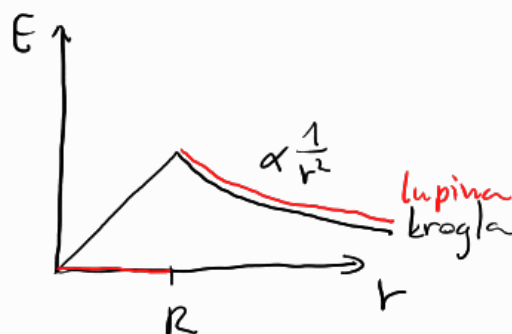
$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = e = \epsilon_0 E 4\pi r^2$$

$$E = \frac{e}{4\pi\epsilon_0 r^2}$$

$r < R:$

$$\epsilon_0 E 4\pi r^2 = e \frac{r^3}{R^3}$$

$$E = \frac{er}{4\pi\epsilon_0 R^3}$$



$$\frac{e(r)}{V(r)} = \frac{e}{V} = \rho_e$$

$$\rightarrow e(r) = e \left(\frac{r}{R} \right)^3$$

$$dV = -\vec{E} \cdot d\vec{r}$$

$r \geq R:$

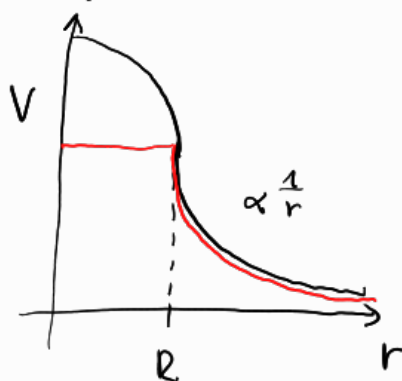
$$\int_{\infty}^{V(r)} dV = - \int_{\infty}^r \vec{E} \cdot d\vec{r} \rightarrow V(r) = + \frac{e}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{\infty} \right) = \frac{e}{4\pi\epsilon_0 r}$$

$\swarrow$
V unendlich
hi potencie

$r \leq R:$

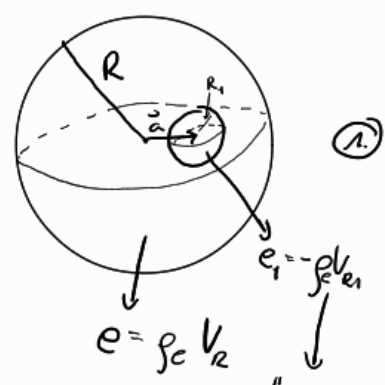
$$\int_{V(R)}^{V(r)} dV = - \int_R^r \vec{E} \cdot d\vec{r} \rightarrow V(r) - \frac{e}{4\pi\epsilon_0 R} = - \frac{e}{4\pi\epsilon_0 R^3} \int_R^r r dr = \frac{e}{4\pi\epsilon_0 R^3} \left(\frac{R^2}{2} - \frac{r^2}{2} \right)$$

$$V(r) = \frac{e}{4\pi\epsilon_0 R} \left(\frac{1}{2} - \frac{r^2}{2R^2} + 1 \right) = \frac{e}{8\pi\epsilon_0 R} \left(3 - \frac{r^2}{R^2} \right)$$



13.1.4

R
 R_1
 $\vec{a}$
 $\vec{E}(r > R) = ?$ ①
 $\vec{E}(\text{znotraj}) = ?$ ②
 $\vec{E}(v \text{ votlini}) = ?$ ③



$\rho_e = \frac{de}{dV} = \frac{3e}{4\pi R^3}$ naboj, če bi bila cela nabita

①
$$\vec{E} = \frac{\rho_e \cancel{4\pi} R^3}{3 \cancel{4\pi} \epsilon_0 r^2} \frac{\vec{r}}{r} - \frac{\rho_e \cancel{4\pi} R_1^3}{3 \cancel{4\pi} \epsilon_0 |\vec{r} - \vec{a}|^3} (\vec{r} - \vec{a}) =$$

$$= \frac{\rho_e}{3 \epsilon_0} \left(\frac{R^3 \vec{r}}{r^3} - \frac{R_1^3 (\vec{r} - \vec{a})}{|\vec{r} - \vec{a}|^3} \right)$$

 "navidezni naboj";
 če ju seštejemo
 dobimo $e = 0$

②
$$\vec{E} = \frac{\rho_e \cancel{4\pi} R^3}{3 \cancel{4\pi} \epsilon_0 R^3} \frac{\vec{r}}{r} - \frac{\rho_e \cancel{4\pi} R_1^3}{3 \cancel{4\pi} \epsilon_0 |\vec{r} - \vec{a}|^3} (\vec{r} - \vec{a}) = \frac{\rho_e}{3 \epsilon_0} \left(\vec{r} - R_1^3 \frac{(\vec{r} - \vec{a})}{|\vec{r} - \vec{a}|^3} \right)$$

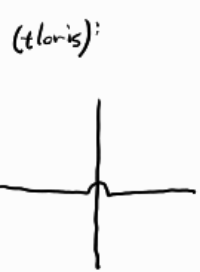
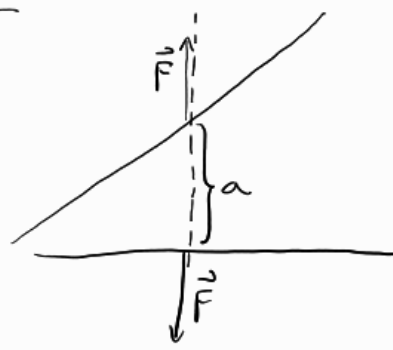
 znotraj velike znotraj male

③
$$\vec{E} = \frac{\rho_e \cancel{4\pi} R^3}{3 \cancel{4\pi} \epsilon_0 R^3} \frac{\vec{r}}{r} - \frac{\rho_e \cancel{4\pi} R_1^3}{3 \cancel{4\pi} \epsilon_0 R_1^3} \frac{(\vec{r} - \vec{a})}{|\vec{r} - \vec{a}|} = \frac{\rho_e}{3 \epsilon_0} (\vec{r} - \vec{r} + \vec{a}) = \frac{\rho_e \vec{a}}{3 \epsilon_0}$$

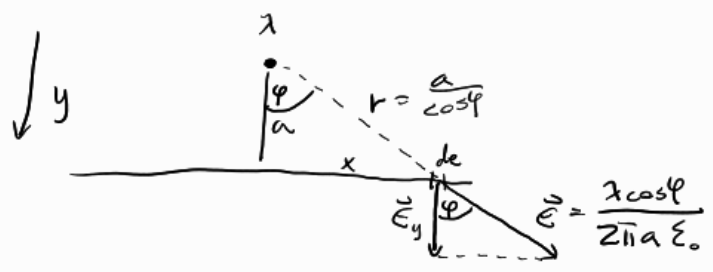
homogeno
polje v smeri a

1991/92/3. kol./2

$a = 5 \text{ cm}$
 $\lambda = 10^{-5} \frac{\text{As}}{\text{m}}$
 $F = ?$



$\vec{F} = e \vec{E}$



$$dF_y = E_y de = \frac{\lambda \cos^2 \varphi}{2\pi a \epsilon_0} de =$$

$$= \frac{\lambda^2}{2\pi \epsilon_0} d\varphi$$

$$F_y = \frac{\lambda^2}{2\pi \epsilon_0} \int_0^{\pi/2} d\varphi = \frac{\lambda^2}{\pi \epsilon_0} \frac{\pi}{2} = \frac{\lambda^2}{2 \epsilon_0}$$

$\vec{E}$ che žice:

$\epsilon_0 \oint \vec{E} d\vec{s} = e$

$\epsilon_0 \cdot E \cdot 2\pi r l = e$

$E = \frac{\lambda}{2\pi \epsilon_0 r}$

$de = \lambda dl$

$dl = dx$

$de = \frac{\lambda a}{\cos^2 \varphi} d\varphi$

$\frac{x}{a} = \tan \varphi$

$dx = a \cdot \frac{1}{\cos^2 \varphi} d\varphi$

95/96/3. kol. / 4

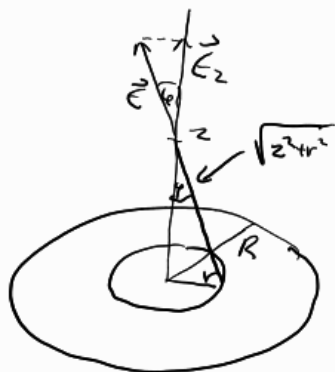
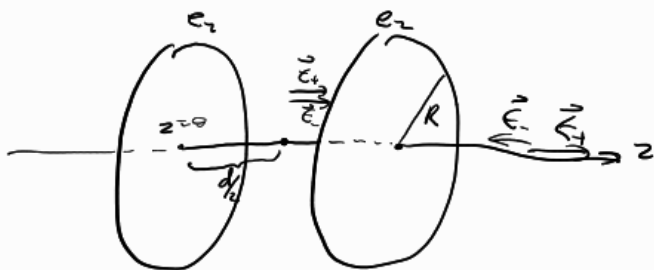
$$R = 2 \text{ cm}$$

$$d = 5 \text{ cm}$$

$$e_1 = -e_2 = 10^{-10} \text{ As}$$

$$E(z, 5) = ?$$

$$E(z \gg d) = ?$$



$$d\vec{E}_z = \frac{de}{4\pi\epsilon_0 \sqrt{z^2 + r^2}} \cdot \cos\varphi = \frac{de \cdot z}{4\pi\epsilon_0 (z^2 + r^2)^{3/2}}$$

$$de = e \cdot \frac{2\pi r dr}{\pi R^2}$$

$$E_z = \frac{e \cdot z}{4\pi\epsilon_0 R^2} \int_0^R \frac{2r dr}{(z^2 + r^2)^{3/2}}$$

$$z^2 + r^2 = u \\ 2r dr = du$$

$$= \frac{e \cdot z}{4\pi\epsilon_0 R^2} \int_{z^2}^{z^2 + R^2} u^{-3/2} du = \frac{e \cdot z}{4\pi\epsilon_0 R^2} 2 \left(\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2 + R^2}} \right) =$$

$$= \frac{e}{2\pi\epsilon_0 R^2} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) = \frac{e}{2\pi\epsilon_0 R^2} \left(1 - \frac{1}{\sqrt{1 + \left(\frac{R}{z}\right)^2}} \right)$$

$$E(z=0) = \frac{Ze}{2\pi\epsilon_0 R^2} \left(1 - \frac{1}{\sqrt{1 + \left(\frac{R}{z}\right)^2}} \right) =$$

$$E(z \gg) = \frac{e}{2\pi\epsilon_0 R^2} \left(1 - \frac{1}{\sqrt{1 + \left(\frac{R}{z}\right)^2}} \right) - \left(1 - \frac{1}{\sqrt{1 + \left(\frac{R}{z-d}\right)^2}} \right) =$$

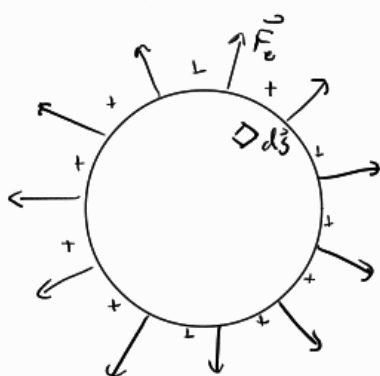
$$x \rightarrow 0 \\ \frac{1}{\sqrt{1+x}} \approx 1 - \frac{1}{2}x$$

$$= \frac{e}{2\pi\epsilon_0 R^2} \left(1 - \frac{1}{2} \frac{R^2}{(z-d)^2} - 1 + \frac{1}{2} \frac{R^2}{z^2} \right) = \frac{e R^2}{4\pi\epsilon_0 R^2} \left(\frac{1}{z^2} - \frac{1}{z^2 (1 - \frac{d}{z})^2} \right) = \underline{\underline{\frac{e a}{2\pi\epsilon_0 z^3}}}$$

13.1.7

$\sigma > 0$

$p = ?$



$$d\vec{F} = \vec{E} de$$

$$p = \frac{dF}{dS} \quad \sigma = \frac{de}{dS}$$

$$p = \frac{E de}{dS} = \sigma E$$

$$E = ?$$

$$p = - \frac{d \left(\frac{e^2}{8\pi\epsilon_0 r} \right)}{4\pi r^2 dr} = \frac{e^2}{8\pi\epsilon_0 \cdot 4\pi r^2} \frac{d \left(\frac{1}{r} \right)}{dr} =$$

$$= \frac{e^2}{8\pi\epsilon_0 \cdot 4\pi r^2} \frac{1}{r^2} = \frac{e^2}{32\pi^2 \epsilon_0 r^4}$$

$$dA = -p dV$$

$$p = - \frac{dA}{dV} = - \frac{dW_e}{dV}$$

$$W_e = \int_0^e \frac{e}{4\pi\epsilon_0 r} de = \frac{1}{2} e \phi V$$

$$dV = 4\pi r^2 dr$$

$$\sigma = \frac{e}{4\pi r^2} \rightarrow E = \frac{e}{8\pi\epsilon_0 r^2} = \frac{\sigma}{2\epsilon_0}$$

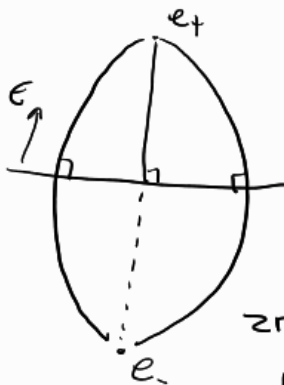
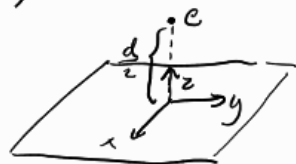
13.1.8

a)

$$a) z = \frac{d}{2}$$

točkast $e > 0$

$$b) z = \frac{d}{2}$$



$$V = \int \vec{E} \cdot d\vec{s} = 0 \rightarrow \vec{E} \perp d\vec{s}$$

potencijal na sp. prevodniku

zrcaljenje
naboja

$$V(x, y, 0) = \frac{e}{4\pi\epsilon_0 \sqrt{(\frac{d}{2})^2 + x^2 + y^2}} - \frac{e}{4\pi\epsilon_0 \sqrt{(\frac{d}{2})^2 + x^2 + y^2}}$$

$$V(x, y, z) = \frac{e}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{x^2 + y^2 + (z - \frac{d}{2})^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z + \frac{d}{2})^2}} \right)$$

$$\frac{\partial V}{\partial z} = \frac{e}{4\pi\epsilon_0} \left(-\frac{1 \cdot z - \frac{d}{2}}{\sqrt{x^2 + y^2 + (z - \frac{d}{2})^2}^3} + \frac{z + \frac{d}{2}}{\sqrt{x^2 + y^2 + (z + \frac{d}{2})^2}^3} \right)$$

$$\vec{E} = -\vec{\nabla} V = \left(-\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y}, -\frac{\partial V}{\partial z} \right) \rightarrow$$

$$E_x = \frac{e}{4\pi\epsilon_0} \left(\frac{x}{\sqrt{x^2 + y^2 + (z - \frac{d}{2})^2}^3} - \frac{x}{\sqrt{x^2 + y^2 + (z + \frac{d}{2})^2}^3} \right)$$

$$E_x(x, y, 0) = 0$$

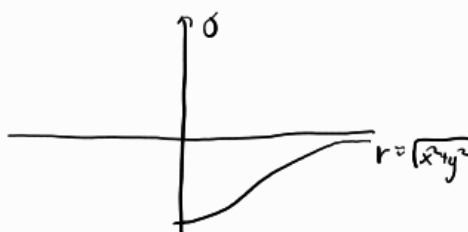
$$E_y(x, y, 0) = 0$$

$$E_z = \frac{e}{4\pi\epsilon_0} \left(\frac{z - \frac{d}{2}}{\sqrt{x^2 + y^2 + (z - \frac{d}{2})^2}^3} - \frac{z + \frac{d}{2}}{\sqrt{x^2 + y^2 + (z + \frac{d}{2})^2}^3} \right)$$

$$E_z(x, y, 0) = -\frac{ed}{4\pi\epsilon_0 \left(x^2 + y^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}}$$

$$\vec{E}(x, y, 0) = (0, 0, E_z)$$

$$\frac{de}{ds} = \sigma = -\epsilon_0 |E| = -\epsilon_0 \frac{ed}{4\pi \left(x^2 + y^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}}$$



b) zica

$$\vec{E}(x, 0, 0) = ?$$

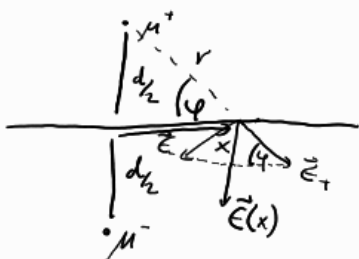
$$r = \sqrt{x^2 + \left(\frac{d}{2} \right)^2}$$

$$\vec{E}_+ = \frac{\mu}{2\pi\epsilon_0 r} \frac{\vec{r}}{r} = \frac{\mu}{2\pi\epsilon_0 \sqrt{x^2 + \left(\frac{d}{2} \right)^2}} \frac{\vec{r}}{r}$$

$$E_z = 2E_+ \sin \varphi = \frac{\mu d}{2\pi\epsilon_0 \left(x^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}}$$

$$\vec{E} = (0, 0, E_z)$$

$$\sigma = -\epsilon_0 E = -\frac{\mu d}{2\pi \left(x^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}}$$



3.1.9

$$r_0 = 10 \text{ cm}$$

$$\varphi = 30^\circ$$

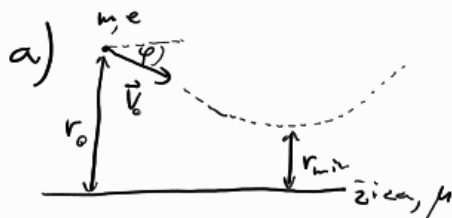
$$\mu = 10^{-6} \frac{\text{As}}{\text{m}}$$

$$e = 10^{-10} \text{ As}$$

$$v_0 = 1 \text{ m/s}$$

$$m = 9,019$$

$$r_{\min} = ?$$



$$E_{z_0} e = \frac{\mu}{2\pi \epsilon_0 r}$$

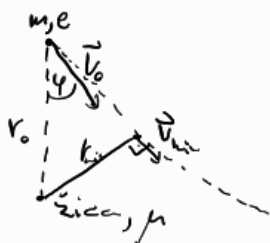
$$F_t = e \frac{\mu}{2\pi \epsilon_0 r} = m \frac{dv_t}{dt} = m v_t \frac{dv_t}{dr}$$

$$\frac{e\mu}{2\pi \epsilon_0} \int_{r_0}^{r_{\min}} \frac{1}{r} dr = m \int_{v_0 \sin \varphi}^0 v_t dv_t \rightarrow \frac{e\mu}{2\pi \epsilon_0} \ln \frac{r_{\min}}{r_0} = -\frac{1}{2} m v_0^2 \sin^2 \varphi$$

$$W_{p,e} = e a z_0 \cos \varphi \quad W_k$$

$$r_{\min} = r_0 \exp\left(-\frac{v_0^2 \sin^2 \varphi \pi \epsilon_0 m}{e\mu}\right) = 5 \text{ cm}$$

b)



$$\Gamma: \Gamma_0 = m r_0 \sin \varphi v_0$$

$$\Gamma_k = m v_{\min} v_{\min}$$

$$v_{\min} = \frac{r_0}{r_{\min}} v_0 \sin \varphi$$

W:

$$W_{p,e0} + W_{k0} = W_{p,en} + W_{kn}$$

$$W_{p,e} = \ln\left(\frac{r_0}{r}\right) \frac{e_1 \mu}{2\pi \epsilon_0}$$

$$\frac{r_0}{r_{\min}} = x$$

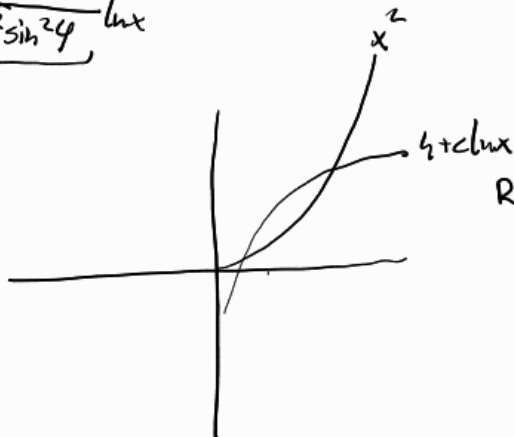
$$m v_0^2 = -\frac{e_1 \mu}{\pi \epsilon_0} \ln x + m v_0^2 \sin^2 \varphi x^2$$

$$x^2 = \frac{1}{\sin^2 \varphi} + \frac{e_1 \mu}{\pi \epsilon_0 m v_0^2 \sin^2 \varphi} \ln x$$

$$x^2 = 4 + c \ln x$$

$$\frac{1}{2} m v_0^2 = \ln\left(\frac{r_0}{r_{\min}}\right) \frac{e_1 \mu}{2\pi \epsilon_0} + \frac{1}{2} m v_{\min}^2$$

$$m v_0^2 = \ln\left(\frac{r_0}{r_{\min}}\right) \frac{e_1 \mu}{\pi \epsilon_0} + m v_0^2 \sin^2 \varphi \left(\frac{r_0}{r_{\min}}\right)^2$$



Resiter: numeričho

$$x \approx \frac{1}{0,56} \rightarrow r_{\min} = 0,56 r_0 = 5,6 \text{ cm}$$



13.1.12

$$a = 2,38 \cdot 10^{-10} \text{ m}$$

$$W = ?$$



$$W_{pe} = e \cdot \left(-2 \frac{e}{4\pi\epsilon_0 a} + 2 \frac{e}{4\pi\epsilon_0 2a} - 2 \frac{e}{4\pi\epsilon_0 3a} + \dots \right) =$$

$$= \frac{e^2}{2\pi\epsilon_0 a} \cdot (-1) \ln 2$$

$$W_{sk} = \frac{N}{2} W_{pe}$$

13.1.13

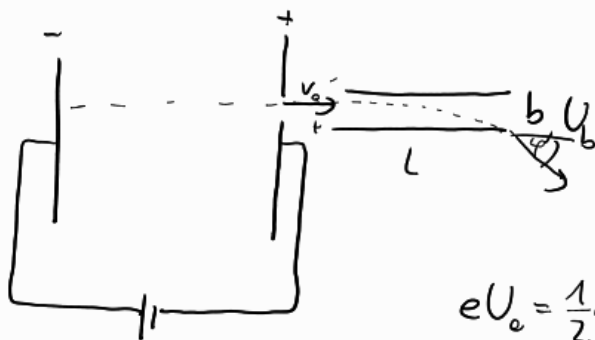
$$U_0 = 3V$$

$$b = 2 \text{ mm}$$

$$l = 1 \text{ cm}$$

$$U_b = 0,2V$$

$$\varphi = ?$$



$$eU_0 = \frac{1}{2} m v_0^2$$

$$e = C U_b = C E b$$

$$U_b = E b$$

$$F_e = eE = \frac{eU_b}{b} = m a_{\perp}$$

$$v_{\perp} = a_{\perp} t = a_{\perp} \frac{l}{v_0} =$$

$$= \frac{e U_b l}{m b v_0}$$

$$\tan \varphi = \frac{v_{\perp}}{v_0} = \frac{e U_b l}{b v_0^2 m} = \frac{e U_b l}{b 2 e U_0} \rightarrow \tan \varphi = \underline{\underline{9,5^\circ}}$$

13.1.15

$$R = 1 \text{ m}$$

$$E_c = 3 \cdot 10^6 \frac{V}{m}$$

(prebojna napetost)

$$e_{max} = ?$$

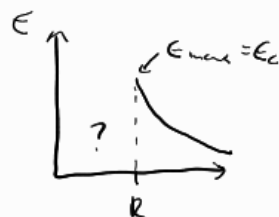
$$U_R = ?$$



$$E = \frac{e}{4\pi\epsilon_0 r^2}$$

$$E_c = \frac{e_{max}}{4\pi\epsilon_0 R^2}$$

$$e_{max} = 3,4 \cdot 10^{-4} \text{ As}$$



$$U = V(R) - V(\infty) = V(R) = \frac{e_{max}}{4\pi\epsilon_0 R} = E_c R = 3 \cdot 10^6 \text{ V}$$

13.1.16

$R, 2R$

e, e

$\Delta e = ?$



$$V = \frac{e}{4\pi\epsilon_0 R}$$

$$\frac{e - \Delta e}{4\pi\epsilon_0 R} = \frac{e + \Delta e}{4\pi\epsilon_0 2R}$$

$$2e - 2\Delta e = e + \Delta e$$

$$\boxed{\Delta e = \frac{e}{3}}$$

13.1.17

Skiciraj silnice $\vec{E}, \vec{V}$

$\vec{E}, \vec{V}$



29/8

$$2r_0 = 1 \text{ mm}$$

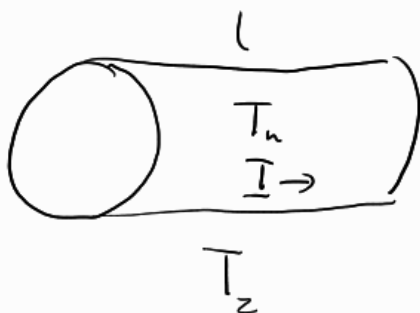
$$T_u = 50^\circ\text{C}$$

$$T_z = 20^\circ\text{C}$$

$$\lambda = 380 \frac{\text{W}}{\text{mK}}$$

$$\xi = 0,017 \Omega \frac{\text{mm}^2}{\text{m}}$$

I_{max}



$$I(r) = I_0 \cdot \frac{S(r)}{S_0} = I_0 \cdot \frac{\pi r^2}{\pi r_0^2}$$

$$P = UI = RI^2 = \int_0^l \frac{1}{S} I^2 = \int_0^l \frac{I(r)^2}{\pi r^2} = \int_0^l \frac{1}{\pi} \frac{I_0^2}{r_0^2} r^2$$

$$P = \lambda \int_0^l S \left(-\frac{dT}{dx} \right) = \lambda \int_0^l 2\pi r l \left(-\frac{dT}{dr} \right)$$

$$\int_0^l \frac{1}{\pi} \frac{I_0^2}{r_0^2} r^2 = \lambda \cdot 2\pi r l \left(-\frac{dT}{dr} \right)$$

$$\int_0^l \frac{I_0^2}{2\pi^2 r_0^2 \lambda} r dr = \int_{T_u}^{T_z} dT \int_0^l \frac{I_0^2}{2\pi^2 r_0^2 \lambda} \frac{r^2}{2} = T_u - T_z$$

$$I_{max} = I_0 = \sqrt{\frac{4\pi^2 r_0^2 \lambda (T_1 - T_2)}{\xi}} = 2\pi \cdot 0,5 \frac{m}{m} \sqrt{\frac{380 W \cdot 30 K \frac{W}{m \cdot K} A}{\frac{W}{m \cdot K} \cdot 0,017 \frac{m}{m^2}}} = 2600 A$$

(29/9)

$$2r_1 = 10 cm$$

$$2r_2 = 25 cm$$

$$\xi = 10^{12} \Omega \frac{mm^2}{m}$$

$$U = 2500 V$$

$$T_2 = 20^\circ C$$

$$\lambda = 0,1 \frac{W}{m \cdot K}$$

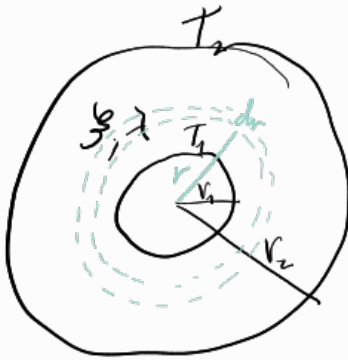
$$I = ?$$

$$T_1 = ?$$

$$I = \frac{U}{R}$$

$$R = \frac{\xi}{4\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$I = 2,7 \cdot 10^{-3} A$$



$$dP = I^2 dR = I^2 \xi \frac{dr}{4\pi r^2}$$

$$P = \frac{I^2 \xi}{4\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \lambda 4\pi r^2 \left(-\frac{dT}{dr} \right)$$

$$\int_{r_1}^{r_2} \frac{I^2 \xi}{4\pi r^2} \left(\frac{1}{r} - \frac{1}{r_0} \right) dr = \lambda 4\pi (T_2 - T_1)$$

r_1

$$\frac{I^2 \xi}{4\pi} \left(\left(\frac{1}{R_1^2} - \frac{1}{R_2^2} \right) \left(-\frac{1}{2} \right) - \frac{1}{R_1} (-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \right) =$$

$$= \frac{I^2 \xi}{4\pi} \left(\frac{1}{2R_2^2} - \frac{1}{2R_1^2} + \frac{1}{R_1^2} - \frac{1}{R_1 R_2} \right) = \lambda 4\pi (T_2 - T_1)$$

(29/11)

$$l = 5 m$$

$$S = 0,1 mm^2$$

$$\xi = 0,5 \Omega \frac{mm^2}{m}$$

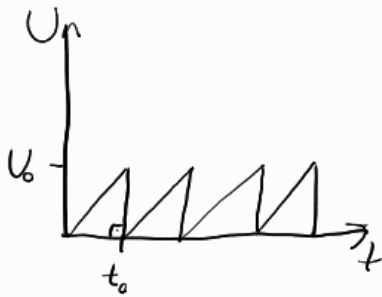
$$c = 450 \frac{J}{kg \cdot K}$$

$$\rho = 89 \frac{kg}{m^3}$$

$$t_0 = 10^{-2} s$$

$$U_0 = 10 V$$

$$\Delta T(t = 1s) = ?$$



$$P = \frac{U^2}{R} = \frac{U^2 S}{\xi l}$$

$$\bar{P} = \frac{\int_0^{t_0} P dt}{t_0} = \frac{S}{\xi l t_0} \int_0^{t_0} U^2 dt =$$

$$= \frac{S U_0^2}{\xi l t_0^3} \int_0^{t_0} t^2 dt = \frac{S U_0^2}{\xi l t_0^3} \frac{t_0^3}{3} = \frac{P_{max}}{3}$$

$$U(t) = \frac{U_0}{t_0} t$$

$$\bar{P} t_1 = mc \Delta T$$

$$\Delta T = \frac{\bar{P} t_1}{c \rho l S} = \frac{P_{max} t_1}{3 \xi l^2 c \rho} = 0,74 K$$

$$C_1 = 0,01 \mu F$$

$$U_1 = 1 \angle 0^\circ$$

$$C_2 = 95 \mu F$$

$$U_2 = 400 \text{ V}$$

a) zvezeno ++, --

b) zvečeno $+$, $+$

$$U_e = ?$$

$$U_b = ?$$

$$\Delta W_{96} = ?$$

a)

The circuit diagram for problem a) shows a voltage source $u_s(t)$ in series with a resistor R_1 . This series combination is connected to a parallel network consisting of a resistor R_2 and an inductor L_1 .

$$|e_v| = |e_1 + e_2| = c_1 v_1 + c_2 v_2$$

$$\Delta W = 0$$

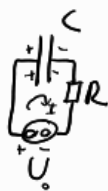
b)

A circuit diagram showing two capacitors connected in series. Each capacitor is represented by two parallel vertical lines. The top plate of the first capacitor is labeled with a minus sign (-) and the bottom plate with a plus sign (+). The top plate of the second capacitor is labeled with a minus sign (-) and the bottom plate with a plus sign (+). The capacitors are connected in a single loop.

$$e_k = |e_2 - e_1| \approx |C_2 V_2 - C_1 V_1|$$

$$\Delta W = W_1 - W_2$$

29/32



$$R = 10^5 \Omega$$

$$C = 10 \mu F$$

$$U_o = kt, U_o(t=0) = 0$$

$t = 1s$

$$\frac{V_R}{U_i}(t_1) = 7$$

$$-IR - V_c + V_o = 0$$

$$L \frac{dI}{dt} - IR - \frac{e}{c} = 0 \quad / \frac{d}{dt}$$

$$k - \frac{dI}{dt} R - \frac{I}{C} = 0$$

$$\frac{dI}{dt}R = L - \frac{I}{C}$$

$$\int_0^1 \frac{dI}{kC-I} = \int_0^t \frac{1}{R_C} dt$$

$$u = L C^{-1}$$

$$du = -dI$$

$$\ln \frac{k_C - I}{k_C} = - \frac{t}{R_C}$$

$$kC - I = kC e^{-\frac{t}{RC}}$$

$$I = I_C (1 - e^{-\frac{t}{R_C}})$$

$$U_R = RI$$

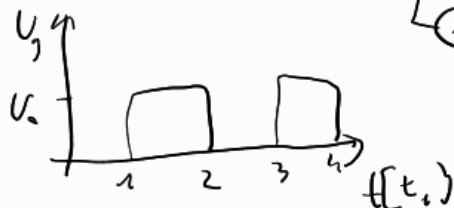
$$U_c = \frac{e}{C}$$

$$Q = \int_0^t I dt =$$

$$= L C \left(t + R C e^{-\frac{t}{RC}} \right) \Big| =$$

$$\frac{U_R}{U_C} = \frac{RkC(1 - e^{-\frac{t}{RC}})}{k(t + RC(e^{-\frac{t}{RC}} - 1))} = kC(t + RC(e^{-\frac{t}{RC}} - 1))$$

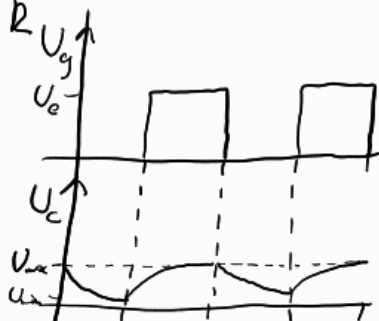
13.3/5



$U_0 = 100V$ $U_c \rightarrow$ periodično

$$t_1 = RC = 10s$$

$$U_{1,0} = U_c(t=0) = ?$$



29/15

$$N=10$$

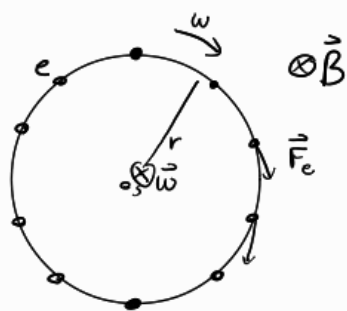
$$e=10^{-3} \text{ As}$$

$$r=10 \text{ cm}$$

$$J=10^{-4} \text{ kg m}^{-2}$$

$$B=0,5 \text{ T}$$

$$\omega(B=0)=?$$



$$M=J\alpha = N F_e r$$

$$\vec{F}_e = e \vec{E}$$

$$\oint \vec{E} \cdot d\vec{s} = U_i = - \frac{d\Phi_m}{dt}$$

$$2\pi r E = - \frac{dB \cdot \pi r^2}{dt}$$

$$E = - \frac{r}{2} \frac{dB}{dt}$$

$$\int_0^\omega J \frac{d\omega}{dt} = \int_0^\omega N e \frac{r^2}{2} \frac{dB}{dt}$$

$$J\omega = B N e \frac{r^2}{2}$$

$$\omega = \frac{B N e r^2}{2J} = 0,25 \text{ s}^{-1}$$

13.6.4



$$f(\omega) = \left| \frac{U_R(\omega)}{U_g(\omega)} \right|$$

$$Z_R = R$$

$$Z_L = i\omega L$$

$$U_g = U_0 e^{i\omega t}$$

$$U_R = RI$$

$$U_g = Z' I$$

$$Z' = Z_R + Z_L = R + i\omega L$$

$$U_0 e^{i\omega t} = |Z'| e^{i\varphi} I$$

$$|Z'| = \sqrt{R^2 + \omega^2 L^2}$$

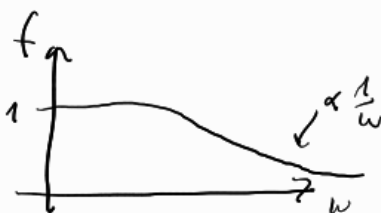
$$I = \frac{U_0 e^{i\omega t}}{|Z'| e^{i\varphi}} =$$

$$\tan(\varphi) = \frac{\omega L}{R}$$

$$= \frac{U_0}{\sqrt{R^2 + \omega^2 L^2}} e^{i(\omega t - \varphi)}$$

gleichen = same amplitude:

$$\left| \frac{U_R}{U_g} \right| = f(\omega)$$



$$Z_R = R$$

$$Z_C = \frac{1}{i\omega C}$$

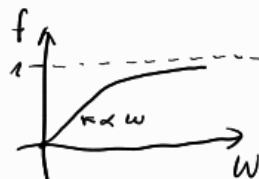
$$Z' = R - i \frac{1}{\omega C}$$

$$|Z'| = \sqrt{R^2 + \frac{1}{\omega^2 C^2}}$$

$$\tan \varphi = \frac{1}{R\omega C}$$

$$I = \frac{U_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} e^{i(\omega t - \varphi)}$$

$$f(\omega) = \left| \frac{U_R}{U_g} \right| = \left| \frac{R}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \right|$$



$$U_R = RI = \frac{U_0 R}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} e^{i(\omega t - \varphi)}$$

13.6.8



$$Z_R = R$$

$$Z_C = \frac{1}{i\omega C}$$

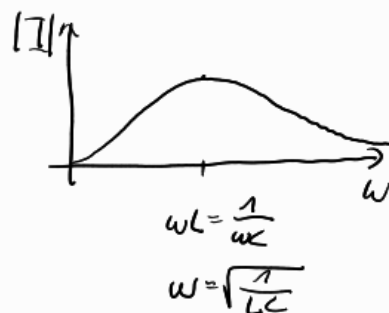
$$Z_L = i\omega L$$

$$Z' = R + i(\omega L - \frac{1}{\omega C})$$

$$|Z'| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$\tan(\varphi) = \frac{\omega L - \frac{1}{\omega C}}{R}$$

$$|I| = \frac{U_0}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$



29/25

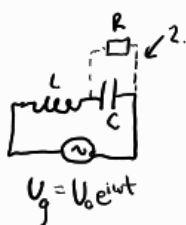
$$\omega = 10^4 \text{ s}^{-1}$$

$$L = 10^{-5} \text{ H}$$

$$C = 50 \mu\text{F}$$

$$R = 1 \Omega$$

$$\frac{I_{u1}}{I_{u2}} = ?$$



$$Z_R = R$$

$$Z_L = i\omega L$$

$$Z_C = -i \frac{1}{\omega C}$$

$$Z_1 = i(\omega L - \frac{1}{\omega C}) \quad |Z_1| = |\omega L - \frac{1}{\omega C}|$$

$$I_1 = \frac{U_0}{Z_1} = \frac{U_0}{|\omega L - \frac{1}{\omega C}|} e^{i(\omega t - \varphi)}$$

$$\tan \varphi_1 = \infty$$

$$\varphi_1 = \frac{\pi}{2}$$

$$Z_2 = i\omega L + \frac{1}{\frac{1}{R} + i\omega C} = i\omega L + \frac{R(1 - iR\omega C)}{(1 + iR\omega C)(1 - iR\omega C)} = i\omega L + \frac{R - iR^2\omega C}{1 + R^2\omega^2 C^2} =$$

$$= \left(\frac{R}{1 + R^2\omega^2 C^2} \right) + i \left(\omega L - \frac{R^2\omega C}{1 + R^2\omega^2 C^2} \right)$$

$$\tan \varphi_2 = \frac{\omega L - \frac{R^2\omega C}{1 + R^2\omega^2 C^2}}{\frac{R}{1 + R^2\omega^2 C^2}} =$$

$$I_2 = \frac{U_0}{\sqrt{\left(\frac{R}{1 + R^2\omega^2 C^2} \right)^2 + \left(\omega L - \frac{R^2\omega C}{1 + R^2\omega^2 C^2} \right)^2}} e^{i(\omega t - \varphi_2)}$$

$$\frac{I_1}{I_2} = \dots = \underline{\underline{0,98}}$$

13.6.6

! Isti podatki

$$U_0 = 100 \text{ V}$$

$$P_{g2} = ?$$

$$P = I_2 U_0 = \frac{U_0}{|Z_2|} \cos(\omega t - \varphi_2) \cdot U_0 \cos(\omega t) =$$

$$= \frac{U_0^2}{|Z_2|} \underbrace{\cos^2(\omega t)}_{\frac{1}{2}} \cos \varphi_2 + \underbrace{\sin \omega t \cos \omega t}_{0} \sin \varphi_2$$

$$\overline{P} = \frac{1}{2} \frac{U_0^2}{|Z_2|} \cos \varphi_2$$

29/5

$$C = 1 \mu F$$

$$L = 10^{-3} H$$

$$t(W_C = \frac{1}{2} W_{C0})$$



$$W_C + W_L = \text{konst.}$$

$$-\frac{e}{C} - L \frac{dI}{dt} = 0 \quad / \frac{d}{dt}$$

$$-\frac{I}{LC} = \frac{d^2 I}{dt^2}$$

$$\omega^2 = \frac{1}{LC}$$

$$I = I_0 \sin(\omega t)$$

$$e = \int I dt = \frac{I_0}{\omega} \cos \omega t$$

$$W_C = \frac{e^2}{2C}$$

$$W_L = \frac{LI^2}{2}$$

$$W_{Cmax} = W_{Lmax} = \frac{LI_0^2}{2}$$

$$\frac{LI_0^2}{2} = \frac{LI_{1/2}^2}{2}$$

$$I_{1/2} = \frac{I_0}{\sqrt{2}} = I_0 \sin(\omega t_{1/2})$$

$$\omega t_{1/2} = \frac{\pi}{4}$$

$$\boxed{\begin{aligned} t_{1/2} &= \frac{1}{\omega} \left(\frac{\pi}{4} + 2k\pi \right) \\ t'_{1/2} &= \frac{1}{\omega} \left(\frac{3\pi}{4} + 2k\pi \right) \end{aligned}} \quad k \in \mathbb{Z}$$

29/29

$$\mu = 500$$

$$S = 10 \text{ cm}^2$$

$$l = 40 \text{ cm}$$

$$N_1 = 1000$$

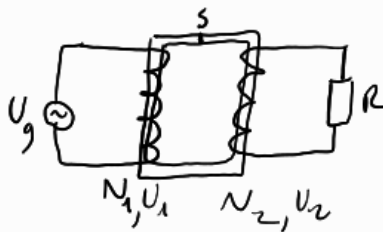
$$N_2 = 400$$

$$R_2 = 500 \Omega$$

$$f = 50 \text{ Hz}$$

$$\sigma(U_1, I_1) = ?$$

$$\frac{I_1}{I_2} = ?$$



$$\frac{U_1}{U_2} = \frac{N_1}{N_2}$$

$$U_1 = U_0 \cos(\omega t)$$

$$Z_2 = R$$

$$\oint \vec{H} \cdot d\vec{s} = \vec{I}$$

$$\frac{l}{\mu \mu_0} B = H l = N_1 I_1 + N_2 I_2$$

$$B = \frac{\mu \mu_0}{l} (I_1 N_1 + I_2 N_2)$$

$$U_1 = - \frac{d\Phi_{m1}}{dt} =$$

$$= - \frac{d}{dt} (BSN_1) = -SN_1 \frac{\mu \mu_0}{l} \frac{d(I_1 N_1 + I_2 N_2)}{dt} = -SN_1 \frac{\mu \mu_0}{l} (\dot{I}_1 N_1 + \dot{I}_2 N_2) = -L_1 \dot{I}_1 - L_{12} \dot{I}_2$$

$$U_2 = -SN_2 \frac{\mu \mu_0}{l} \frac{d(I_1 N_1 + I_2 N_2)}{dt} = -SN_2 \frac{\mu \mu_0}{l} (\dot{I}_1 N_1 + \dot{I}_2 N_2) = -(L_{12} \dot{I}_1 + L_2 \dot{I}_2)$$

$$U_1 + U_2 = 0$$

$$U_2 - I_2 R = 0$$

$$U_g = L_1 \dot{I}_1 + L_{12} \dot{I}_2$$

$$I_2 R = -L_{12} \dot{I}_1 - L_2 \dot{I}_2$$

↓

$$U_g = i\omega I_1 L_1 + i\omega I_2 L_{12}$$

$$I_2 R = -i\omega I_1 L_{12} - i\omega I_2 L_2 \rightarrow I_2 (R + i\omega L_2) = -i\omega I_1 L_{12} \rightarrow I_2 = -\frac{i\omega I_1 L_{12}}{R + i\omega L_2}$$

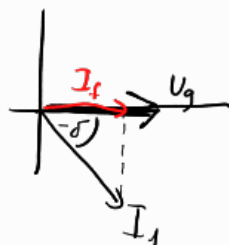
$$U_g = i\omega I_1 \left(L_1 + \frac{\omega^2 L_{12}^2 (R - i\omega L_2)}{(R + i\omega L_2)(R - i\omega L_2)} \right) = \omega I_1 \left(iL_1 + \frac{\omega R L_{12}^2}{R^2 + \omega^2 L_2^2} - i \frac{\omega^2 L_{12}^2 L_2}{R^2 + \omega^2 L_2^2} \right)$$

$$Z = \frac{\omega^2 R L_{12}^2}{R^2 + \omega^2 L_2^2} + i \left(\frac{\omega L_1 R^2 - \omega^3 L_1 L_2^2}{R^2 + \omega^2 L_2^2} \right) = \frac{R \omega^2 L_{12}^2}{R^2 + \omega^2 L_2^2} + i \frac{\omega L_1 R^2}{R^2 + \omega^2 L_2^2}$$

$$|Z| = \frac{\sqrt{R^2 \omega^4 L_{12}^4 + \omega^2 L_1^2 R^4}}{R^2 + \omega^2 L_2^2} = \frac{R L_1 \omega \sqrt{\omega^2 L_2^2 + R^2}}{R^2 + \omega^2 L_2^2} = \frac{R L_1 \omega}{\sqrt{R^2 + \omega^2 L_2^2}}$$

$$\tan \delta = \frac{\text{Im}}{\text{Re}} = \frac{\omega L_1 R^2}{R \omega^2 L_{12}^2} = \frac{R}{\omega L_2}$$

$$I_1 = \frac{U_g}{Z} = \frac{U_0}{|Z|} e^{i(\omega t - \delta)} \rightarrow \delta = 79^\circ$$

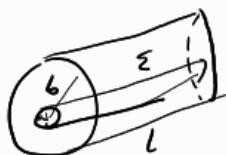


$$I_f = I_1 \cos \delta$$

$$\frac{I_1}{I_f} = \frac{1}{\cos \delta} = 1.7$$

14.1.1

$\frac{C}{l} = ?$



$$\frac{C}{l} = \frac{2\pi \epsilon \epsilon_0}{\ln \frac{a}{b}}$$

$\frac{L}{l} = ?$

$$\frac{L}{l} = \frac{\mu_0 \ln \frac{b}{a}}{2\pi}$$

Impedance koaks. kabla (samo za val v eno smer)

$$Z_k = \sqrt{\frac{R + i\omega L}{G + i\omega C}} = \frac{U_{\rightarrow}}{I_{\rightarrow}} = \frac{U_{\leftarrow}}{I_{\leftarrow}}$$

$R \rightarrow$ upornost

$G \rightarrow$ prevodnost

$$U = (U_{0\rightarrow} e^{i(\omega t - kx)} e^{-\beta x} + U_{0\leftarrow} e^{i(\omega t + kx)} e^{\beta x}) e^{-\beta_c t}$$

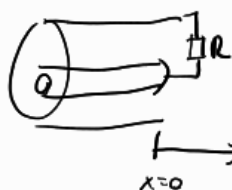
$$I = (I_{0\rightarrow} e^{i(\omega t - kx)} e^{-\beta x} - I_{0\leftarrow} e^{i(\omega t + kx)} e^{\beta x}) e^{-\beta_c t}$$

$$L_{12}^2 = L_1 L_2$$

14.1.5

a, b, ε

$U_{0\rightarrow}, R, \omega$



$$x=0 \rightarrow U_R = U_{0\rightarrow} e^{i\omega t} + U_{0\leftarrow} e^{i\omega t} = (U_{0\rightarrow} + U_{0\leftarrow}) e^{i\omega t}$$

$$I_R = (I_{0\rightarrow} - I_{0\leftarrow}) e^{i\omega t}$$

$$R = \frac{U_R}{I_R} = \frac{U_{0\rightarrow} + U_{0\leftarrow}}{I_{0\rightarrow} - I_{0\leftarrow}}$$

$$Z_k = \sqrt{\frac{L}{C}} = \frac{U_{0\rightarrow}}{I_{0\rightarrow}} = \frac{U_{0\leftarrow}}{I_{0\leftarrow}}$$

$$R(U_{0\rightarrow} - U_{0\leftarrow}) = Z_k(U_{0\rightarrow} + U_{0\leftarrow})$$

$$U_{0\leftarrow} = U_{0\rightarrow} \frac{R - Z_k}{R + Z_k} \quad d)$$

a) $U_{0\leftarrow} = -U_{0\rightarrow} ; \delta = 180^\circ$

b) $U_{0\leftarrow} = U_{0\rightarrow} ; \delta = 0$

c) $R = Z_k$

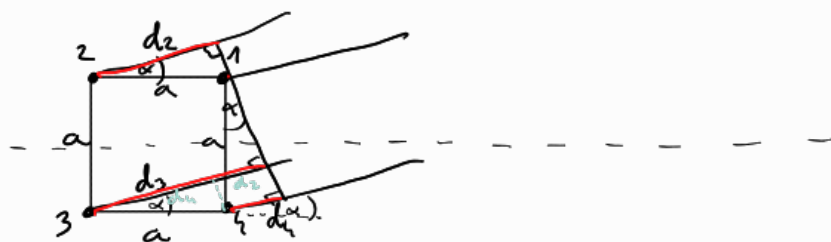
29/21

$a = 10m$

$\delta = 0$

$V = ?$

(najv. 3 pojačanja)



$$d_2 = N_2 \lambda = a \cos \alpha$$

$$d_3 = N_3 \lambda = a (\sin \alpha + \cos \alpha)$$

$$d_4 = N_4 \lambda = a \sin \alpha$$

$$\rightarrow \tan \alpha = \frac{N_4}{N_2}$$

N_2	N_4	α	λ	V
1	0	0	a	a/a
2	0	0	$a/2$	$2a/a$
1	1	$\pi/4$	$a/\sqrt{2}$	$\sqrt{2}a/a$
2	1	$26,5^\circ$	$0,45a$	
			$\frac{1}{2}a$	

29/2

$$n_v = 1,33$$

$$d = 0,5 \mu m$$

$$n_o = 1,4$$

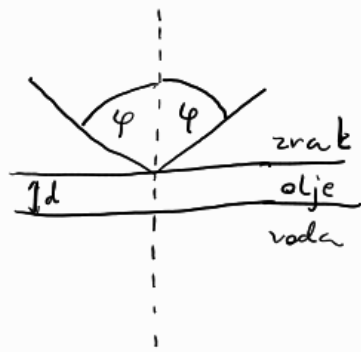
$$\lambda_1 = 500 nm$$

$$\lambda_2 = 620 nm$$

$$\varphi_3 = 45^\circ$$

$$\varphi_{1,2} = ?$$

$$\lambda_3 = ?$$



$n_1 < n_2$	$n_1 > n_2$
Faza se obrće	Faza se <u>ne</u> obrće